

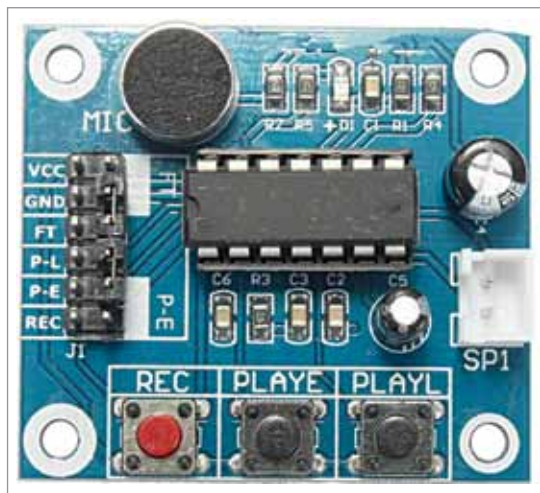


Fig. 3: ISD1820 voice recorder/playback module



Fig. 4: TDA2030A 18W audio amplifier module



Fig. 5: Single-changeover relay module

of ISD1820 module. A 12V DC power supply is connected to this module and Arduino board.

It is a single-channel 18W power amplifier circuit designed around IC TDA2030A. The 10k adjustable resistance works as volume control. Speaker LS2 is connected at the output as shown.

The commonly used 220V AC school bell or buzzer can be connected to the relay's normally-open (NO) pins. The 5V, single-changeover relay used for the purpose is shown in Fig. 5.

Another important module, which acts as an input to the microcontroller, is the DS1307 RTC module shown in Fig. 6. It keeps the clock running. The DS1307 serial real-time clock (RTC) is a low-power device with a 3V CR2032 lithium back-up battery. Its data (seconds, minutes, hours, day, date, month, and year information) is transferred serially to microcontroller through an I2C bidirectional bus. Apart from the power supply, only SDA (data line) and SCL (clock line) pins are required for establishing

the I2C communication. The pin connections between I2C module and the Arduino board are shown in the circuit diagram.

Table 1 shows the pin connections between Arduino Uno board and other modules in the project. Arduino Uno is a microcontroller board based on 8-bit ATmega328P microcontroller that is simple to use and interface. Mini breadboard can be used for power supply connections between the modules and Arduino Uno board.

Software

The circuit operates through software program schoolbell.ino loaded into the internal memory of Arduino Uno. The program implements all the necessary functionalities. It is simple and easy to understand. Comments are given at the end of each command line.

In this project the following two header files are necessary in the main code:

#include <Wire.h>. The Wire library allows to communicate through I2C devices, also called 2wire or TWI (TwoWireInterface). The SDA (data line) and SCL (clock line) are used for communicating with DS1307 RTC module.

#include "RTCLib.h". Used For communicating with DS1307 RTC module.

In the next step, the code initialises the relay and ISD1820 module P-L pin as output. Then, *Serial.begin(9600);* command is used for serial communication for displaying the time on the serial monitor screen. It is mainly used for reference purpose.

In the loop, set the required timings for the alert message and the relay (buzzer) on and off timings. Two methods of setting the time in the RTC module are given below:

Method 1. *rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));* line sets

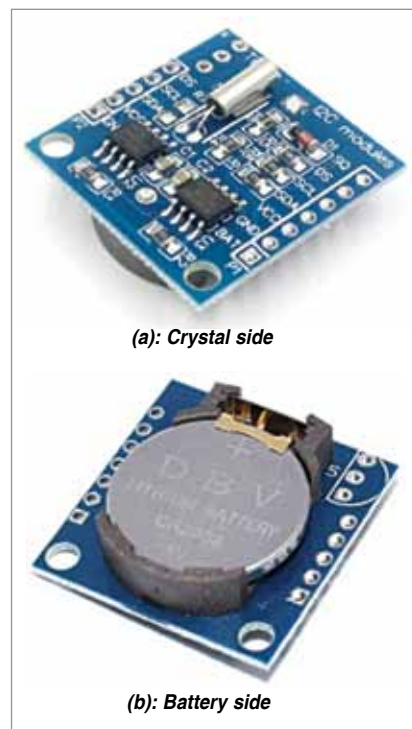


Fig. 6: DS1307 I2C RTC module

Arduino Pins' and Modules' Connections

Table 1

Pin	Connection
5V DC	Connected to Vcc pin of ISD1820 module
5V DC	Connected to Vcc pin of 5V relay module
5V DC	Connected to Vcc pin of DS1307 RTC module
12V DC	Connected to Vcc pin of TDA2030A amplifier module
A4	SDA pin of DS1307 RTC module
A5	SCL pin of DS1307 RTC module
Pin 11	IN pin of relay board
Pin 12	P-L pin of ISD1820 module
GND	Connected GND pin of relay board
GND	Connected to GND pin of ISD1820 module
GND	Connected to GND pin of DS1307 RTC module
GND	Connected to GND pin of TDA2030A amplifier module